The Savage–Dickey Test for Zero Migration in BA3

BA3 Technical Documentation

BA3 3.6.0 and later; option `-B` from 3.7.0

Abstract

For every ordered pair of populations, BA3 reports a Bayes factor for the hypothesis of no migration, $H_0 : m_{ij} = 0$, against $H_1 : m_{ij} > 0$, computed by the Savage–Dickey density ratio. This note gives the model, the derivation and the implementation used from BA3 3.6.0 onward. Because the migration rates affect the data only through the ancestry of the sampled individuals, the posterior density of m_{ij} at zero can be computed exactly given the sampled ancestries, and averaged over the MCMC sample (Rao–Blackwellization). The result is

$$\mathrm{BF}_{01} \;=\; \Pr(c_{ij} = 0 \mid \mathcal{D})\,\frac{n_i + K - 1}{K - 1},$$

where c_{ij} is the number of individuals sampled in population i that are first- or second-generation migrants from j, n_i is the number of individuals sampled in i, and K is the number of populations. The estimate is exact up to Monte Carlo error: there is no bandwidth or threshold to choose. It shows that the evidence for H_0 can never exceed $(n_i + K - 1)/(K - 1)$, so it grows with the number of individuals sampled, not with the number of loci.

By default $\Pr(c_{ij} = 0 \mid \mathcal{D})$ is estimated by the fraction of MCMC states with no migrant on the route. When migration is strongly supported no state has $c_{ij} = 0$, and only a resolution limit can be printed. The option `-B` (Section 6) instead averages, over the states, the exact probability that no sampled individual is a migrant given the state's migration rates, allele frequencies and inbreeding coefficients. This estimate is positive on every route, so BF_{01} is estimated however strong the evidence for migration. Together the two results show that migration is far easier to detect than to exclude: a single clearly identified migrant can make the evidence for migration overwhelming, while the evidence against it is capped by the number of individuals sampled. Section 12 describes the kernel-density estimator used in versions 3.4.0–3.5.0 and the two errors that made it overstate BF_{01}.

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1 The migration model and its prior

1.1 Migration rates and ancestry

Let K be the number of sampled populations. For a receiving population i, m_{ij} ($j \neq i$) is the fraction of individuals in i that are migrants, or offspring of migrants, from j within the last two generations, and $m_{ii} = 1 - \sum_{j \neq i} m_{ij}$. An individual sampled in i has one of $2K - 1$ ancestries, with prior probabilities

$$\Pr(\text{non-migrant}) = 1 - 3 \sum_{j \neq i} m_{ij}, \tag{1}$$

$$\Pr(\text{first-generation migrant from } j) = m_{ij}, \tag{2}$$

$$\Pr(\text{second-generation migrant from } j) = 2\, m_{ij}. \tag{3}$$

These probabilities must be non-negative, which requires

$$m_{ij} \geq 0, \qquad \sum_{j \neq i} m_{ij} \leq \tfrac{1}{3} \quad (\text{equivalently } m_{ii} \geq \tfrac{2}{3}). \tag{4}$$

For the n_i individuals sampled in population i, let z denote their ancestries and define the counts

$$\begin{aligned} n_{i0} &= \text{number of non-migrants}, \\ n_{ij1} &= \text{number of first-generation migrants from } j, \quad j \neq i, \\ n_{ij2} &= \text{number of second-generation migrants from } j, \quad j \neq i, \end{aligned}$$

and the total number of migrants from j, of either generation,

$$c_{ij} = n_{ij1} + n_{ij2}, \tag{5}$$

so that $n_{i0} + \sum_{j \neq i} c_{ij} = n_i$. The event $c_{ij} = 0$ (the sample from i contains no migrants from j) is the sample counterpart of $H_0 \colon m_{ij} = 0$ in Section 2. In the source code the counts are `migrantCounts[i][i][0]`, `migrantCounts[i][j][1]` and `migrantCounts[i][j][2]`.

1.2 The prior on migration rates

The rows of the migration matrix are a priori independent, and each row is uniform on the region (4). With the change of variables

$$u_{ij} = 3\, m_{ij} \quad (j \neq i), \qquad u_{i0} = 1 - \sum_{j \neq i} u_{ij},$$

the region becomes the $(K-1)$-dimensional probability simplex in the K coordinates $(u_{i0}, u_{ij}, j \neq i)$, and the uniform prior is

$$(u_{i0}, u_{ij}, j \neq i) \sim \text{Dirichlet}(1, \ldots, 1) \quad \text{over } K \text{ categories.} \tag{6}$$

Derivation. The free coordinates of row i are the $d = K - 1$ off-diagonal values u_{ij}, $j \neq i$; u_{i0} is determined by them. The map $m \mapsto u = 3m$ has constant Jacobian 3^d, so a prior that is uniform in m on (4) is uniform in u on the region

$$S_d(1) = \Big\{ u_{ij} \geq 0,\ \textstyle\sum_{j \neq i} u_{ij} \leq 1 \Big\}.$$

A corner simplex $S_d(r) = \{x \in \mathbb{R}^d : x_k \geq 0,\ \sum_k x_k \leq r\}$ has volume $r^d/d!$. The uniform density on $S_d(1)$ is therefore $d! = (K-1)!$, which is the Dirichlet$(1, \ldots, 1)$ density over K categories, $\Gamma(K)/\Gamma(1)^K = (K-1)!$. This proves (6).

To obtain the marginal of one coordinate, fix $u_{ij} = u \in [0, 1]$. The other $d - 1 = K - 2$ off-diagonal coordinates range over $S_{K-2}(1-u)$, whose volume is $(1-u)^{K-2}/(K-2)!$. Integrating them out,

$$p(u_{ij} = u) = (K-1)!\, \frac{(1-u)^{K-2}}{(K-2)!} = (K-1)(1-u)^{K-2}, \qquad 0 \leq u \leq 1,$$

which is the Beta$(1, K-1)$ density. Finally, $m_{ij} = u_{ij}/3$ gives $p(m_{ij} = m) = 3\, p(u_{ij} = 3m)$, so the marginal prior of m_{ij} is

$$p(m_{ij}) = 3(K-1)\,(1 - 3m_{ij})^{K-2}, \qquad 0 \leq m_{ij} \leq \tfrac{1}{3}, \tag{7}$$

and setting $m_{ij} = 0$,

$$\boxed{p(m_{ij} = 0) = 3(K-1).} \tag{8}$$

For $K = 2$ there is a single off-diagonal rate, S_0 is a point, and (7) reduces to the Uniform$(0, \frac{1}{3})$ density 3. For $K > 2$ the marginal piles up toward zero because the $K - 1$ off-diagonal rates share the total of at most $\frac{1}{3}$.

The prior the sampler targets. Equation (6) is the prior the MCMC actually samples. The migration-rate move in `src/main.cpp` picks a row i and one source j, proposes $m'_{ij} = m_{ij} + \delta(U - \frac{1}{2})$ with $U \sim U(0, 1)$, and reflects m'_{ij} into $[0, \frac{1}{3} - \sum_{k \neq i,j} m_{ik}]$. For fixed values of the other rates, this proposal is symmetric. The Metropolis–Hastings ratio contains only the probability of the ancestry counts under (1)–(3), with no prior term. The stationary distribution is therefore the likelihood times a prior that is flat on (4). As a check, running this move with all counts set to zero (so that the chain samples the prior) gives $\Pr(m_{ij} < \varepsilon)/\varepsilon$ with $\varepsilon = 0.002$ equal to 9.0 for $K = 4$ and 26.6 for $K = 10$. The exact values are 8.9 and 26.4, and (8) gives 9 and 27 in the limit.

1.3 Prior probability of no migrants in the sample

Each of the n_i individuals is, independently given the migration rates, a migrant from j (of either generation) with probability $m_{ij} + 2m_{ij} = 3m_{ij} = u_{ij}$, so given u_{ij} the probability that none is a migrant from j is $(1 - u_{ij})^{n_i}$. Averaging over the marginal prior $(K-1)(1-u)^{K-2}$ of u_{ij} derived above,

$$\Pr(c_{ij} = 0) = \mathbb{E}\big[(1 - u_{ij})^{n_i}\big] = \int_0^1 (K-1)(1-u)^{n_i+K-2}\, du = \frac{K-1}{n_i + K - 1}. \tag{9}$$

(More generally, c_{ij} is beta-binomial with parameters $(n_i, 1, K-1)$.) This is the prior probability that none of the n_i sampled individuals is a migrant from j.

2 The Savage–Dickey density ratio

2.1 Hypotheses

For each ordered pair (i, j), $j \neq i$, BA3 tests

$$\begin{aligned} &H_0 \colon m_{ij} = 0 \quad \text{(no migration from } j \text{ into } i\text{)}, \\ &H_1 \colon m_{ij} > 0, \end{aligned}$$

where H_1 is the full BA3 model. The Bayes factor is $\mathrm{BF}_{01} = p(\mathcal{D} \mid H_0)/p(\mathcal{D} \mid H_1)$.

2.2 The identity

When H_0 fixes a parameter at a point inside the parameter space of H_1, and the prior on the remaining parameters under H_0 equals their conditional prior under H_1 at that point, then (Dickey, 1971; Verdinelli & Wasserman, 1995)

$$\mathrm{BF}_{01} = \frac{p(m_{ij} = 0 \mid \mathcal{D}, H_1)}{p(m_{ij} = 0 \mid H_1)}. \tag{10}$$

The ratio is the posterior density at zero over the prior density at zero, both under the full model.

The condition holds here. Under H_1, the conditional prior of the remaining coordinates of row i given $u_{ij} = 0$ is Dirichlet$(1, \ldots, 1)$ over the other $K - 1$ categories. That is, it is flat on the reduced region. All other parameters (other rows, allele frequencies, inbreeding coefficients and, with `-H`, the ancestral frequencies and drift parameters) are a priori independent of row i. H_0 is therefore the BA3 model with the route $j \to i$ removed and the same flat prior on everything else, which is the natural null model.

3 Exact posterior density at zero

3.1 The full conditional of a migration row

The data $\mathcal{D}$ (genotypes) depend on the migration rates only through the ancestries z: given z, the genotype probabilities involve allele frequencies and inbreeding coefficients but not m. Hence $m \perp \mathcal{D} \mid z$, and the full conditional of row i depends only on the counts of that row. Under (1)–(3), the probability of the ancestries of the n_i individuals is

$$p(z_i \mid u) \;=\; u_{i0}^{n_{i0}} \prod_{j \neq i} \Big(\frac{u_{ij}}{3}\Big)^{n_{ij1}} \Big(\frac{2u_{ij}}{3}\Big)^{n_{ij2}} \;=\; u_{i0}^{n_{i0}} \prod_{j \neq i} \frac{2^{\,n_{ij2}}}{3^{\,c_{ij}}}\, u_{ij}^{c_{ij}} \;\propto\; u_{i0}^{n_{i0}} \prod_{j \neq i} u_{ij}^{c_{ij}},$$

using $u_{i0} = 1 - \sum_{j \neq i} u_{ij}$ and (5). Each migrant from j contributes one factor of u_{ij}, whichever its generation. The factor 2 for a second-generation migrant does not involve u and cancels on normalization, so only the total c_{ij} matters. With the Dirichlet prior (6),

$$(u_{i0}, u_{ij}, j \neq i) \mid z \;\sim\; \text{Dirichlet}\big(n_{i0} + 1,\; c_{ij} + 1,\; j \neq i\big), \tag{11}$$

whose parameters sum to $n_i + K$. The marginal of one coordinate is

$$u_{ij} \mid z \;\sim\; \text{Beta}\big(c_{ij} + 1,\; n_i + K - 1 - c_{ij}\big). \tag{12}$$

3.2 Density at zero

A Beta(a, b) density at zero is b if $a = 1$ and 0 if $a > 1$. From (12), and multiplying by 3 for $m_{ij} = u_{ij}/3$,

$$p(m_{ij} = 0 \mid z) = \begin{cases} 3\,(n_i + K - 1) & \text{if } c_{ij} = 0, \\ 0 & \text{if } c_{ij} \geq 1. \end{cases} \tag{13}$$

If even one sampled individual is a migrant from j, a migration rate of exactly zero is impossible. If none is, the density at zero is the largest value a Beta$(1, \cdot)$ density can take for this sample size.

3.3 Rao–Blackwellization

The posterior density at zero is the posterior expectation of the conditional density (Gelfand & Smith, 1990):

$$p(m_{ij} = 0 \mid \mathcal{D}) = \mathbb{E}\big[p(m_{ij} = 0 \mid z) \,\big|\, \mathcal{D}\big] = 3\,(n_i + K - 1)\,\Pr(c_{ij} = 0 \mid \mathcal{D}). \tag{14}$$

Dividing by the prior density (8) gives the Bayes factor. The factor 3 from the Jacobian of $m = u/3$ cancels.

Proposition 1. *Under the BA3 model,*

$$\boxed{\text{BF}_{01} = \Pr(c_{ij} = 0 \mid \mathcal{D})\,\frac{n_i + K - 1}{K - 1} = \frac{\Pr(c_{ij} = 0 \mid \mathcal{D})}{\Pr(c_{ij} = 0)}.} \tag{15}$$

The second form, from (9), shows that BF_{01} is the ratio of posterior to prior probability that the sample from i contains no migrants from j.

4 Properties

Uninformative data. If the genotypes carry no information about ancestry, then $\Pr(c_{ij} = 0 \mid \mathcal{D}) = \Pr(c_{ij} = 0)$ and $\text{BF}_{01} = 1$.

Evidence for no migration is bounded. Since $\Pr(c_{ij} = 0 \mid \mathcal{D}) \leq 1$,

$$\text{BF}_{01} \leq \frac{n_i + K - 1}{K - 1}. \tag{16}$$

The bound is reached when the data establish with certainty that no sampled individual is a migrant from j. It cannot be exceeded however many loci are typed: perfect knowledge of the ancestry of n_i individuals still cannot exclude a small positive m_{ij}. More loci bring $\Pr(c_{ij} = 0 \mid \mathcal{D})$ closer to 0 or 1, and only more individuals raise the ceiling. Table 1 gives values. On the Jeffreys scale (Table 2), "strong" evidence for H_0 ($\text{BF}_{01} > 10$) needs $n_i > 9(K-1)$, and "decisive" evidence ($\text{BF}_{01} > 100$) needs $n_i > 99(K-1)$.

	n_i				
K	10	20	50	100	200
2	11.0	21.0	51.0	101.0	201.0
4	4.3	7.7	17.7	34.3	67.7
10	2.1	3.2	6.6	12.1	23.2

Table 1: Largest attainable BF_{01}, $(n_i + K - 1)/(K - 1)$.

Evidence for migration is unbounded. $\mathrm{BF}_{01} \to 0$ as $\Pr(c_{ij} = 0 \mid \mathcal{D}) \to 0$, so the evidence for migration can be arbitrarily strong. The default estimator cannot measure it once $\Pr(c_{ij} = 0 \mid \mathcal{D})$ is below about $1/N$ for N recorded states (Section 5); the estimator of Section 6 (`-B`) can.

Migration is easier to detect than to exclude. The two previous properties are asymmetric. Evidence for migration on a route needs only one sampled individual whose genotypes clearly place it as a migrant or the offspring of one; evidence against migration needs many individuals, none of which is a migrant, and is capped by (16) however many loci are typed. In the Malagasy palm data set BN (52 individuals from 8 populations, 6–7 each, 5,524 SNPs; `-c -H`), no absent route can exceed $\mathrm{BF}_{01} = (7 + 7)/7 = 2.0$, and all 53 are "Weak for H_0". One individual in population BN4, with probability 1 a second-generation migrant from BN3, gives $\log_{10} \mathrm{BF}_{01} \approx -176$ for that route with `-B`, although m is barely estimated (0.05 ± 0.03). Decisive evidence for migration is evidence under the BA3 model: in the same data, 6 of the 7 individuals in BN1 are assigned as second-generation migrants from BN2 ($\log_{10} \mathrm{BF}_{01} \approx -1031$), a pattern that can also arise when two samples are not distinct populations.

Relation to interval-null estimators. A common way to estimate (10) from MCMC output is to replace the point null by a small interval:

$$\mathrm{BF}_{01} \approx \frac{\Pr(m_{ij} < \varepsilon \mid \mathcal{D})}{\Pr(m_{ij} < \varepsilon)}.$$

This is the Bayes factor for $m_{ij} < \varepsilon$ against the full model, and it converges to (10) as $\varepsilon \to 0$. Given z, both terms are known exactly: $\Pr(m_{ij} < \varepsilon \mid z) = 1 - (1 - 3\varepsilon)^{n_i + K - 1}$ when $c_{ij} = 0$, and $\Pr(m_{ij} < \varepsilon) = 1 - (1 - 3\varepsilon)^{K-1}$. Equation (15) is the $\varepsilon \to 0$ limit taken analytically, so it has neither the bias of a finite ε nor the extra variance of counting raw samples of m_{ij} near zero. As a check, the BA3 migration move run with fixed counts ($n_i = 50$, $K = 10$, $c_{ij} = 0$) gave $\Pr(m_{ij} < 0.001)/0.001 = 162$, equal to the exact finite-ε value 162, which tends to $3(n_i + K - 1) = 177$ as $\varepsilon \to 0$.

Model misspecification. The test is exact for the BA3 model. It inherits that model's failures. In particular, if two samples come from one randomly mating population, BA3 can only explain their shared allele frequencies by migration, so $c_{ij} > 0$ in every sampled state and BF_{01} is reported as decisive for migration in both directions (see the manual, "Populations that should be merged").

5 Monte Carlo estimation

5.1 Estimator

After burn-in, every s iterations (the sampling interval `-n`), BA3 records for each route whether $c_{ij} = 0$ in the current state. With N recorded states $z^{(1)}, \ldots, z^{(N)}$,

$$\widehat{\Pr}(c_{ij} = 0 \mid \mathcal{D}) = \frac{1}{N} \sum_{t=1}^{N} \mathbf{1}\big[c_{ij}^{(t)} = 0\big], \qquad \widehat{\mathrm{BF}}_{01} = \widehat{\Pr}(c_{ij} = 0 \mid \mathcal{D}) \, \frac{n_i + K - 1}{K - 1}. \tag{17}$$

The Monte Carlo standard error of $\widehat{\Pr}$ is approximately $\sqrt{\hat{p}(1-\hat{p})/N_{\mathrm{eff}}}$, where N_{eff} is the effective sample size of the indicator series. Its relative error is largest when $\hat{p}$ is small, which is when the evidence for migration is strong.

5.2 When no recorded state has $c_{ij} = 0$

If $\widehat{\Pr} = 0$, the estimate of BF_{01} would be 0. BA3 instead reports

$$\mathrm{BF}_{01} < \frac{1}{N} \, \frac{n_i + K - 1}{K - 1},$$

the value that a single recorded state with $c_{ij} = 0$ would give, marked with `<`. The interpretation gets the suffix "or more" (for example, "Strong for H_1 or more") unless it is already "Decisive for H_1". This is a reporting convention, not a guaranteed bound: it says only that the run was too short to resolve values below about $1/N$. Even for N independent states, zero occurrences give $\Pr(c_{ij} = 0 \mid \mathcal{D}) < 1/N$ only with about 63% confidence ($1 - e^{-1}$); a 95% limit is about $3/N$. The printed value, and with it the interpretation, therefore depends on the run length and the sampling interval, not on the data. The estimator of Section 6 (option `-B`) gives a positive estimate on such routes.

5.3 Other details

Routes with fewer than 100 recorded states are reported as "Insufficient samples". The estimate depends on the sampler mixing between states with $c_{ij} = 0$ and $c_{ij} > 0$. A route whose indicator never or rarely changes during a run, or whose $\widehat{\Pr}$ differs between seeds, should be treated with caution. The collapsed sampler (`-c`) and the hierarchical allele-frequency prior (`-H`) change how the ancestries z are updated and what prior the allele frequencies have, but not the prior on m or the migration move, so (15) applies unchanged.

6 Exact estimates for strongly supported routes (`-B`)

6.1 The estimator

The counting estimator of Section 5 averages the indicator $\mathbf{1}[c_{ij} = 0]$ over the recorded states. When migration on a route is strongly supported, $\Pr(c_{ij} = 0 \mid \mathcal{D})$ is far below $1/N$ and the indicator is almost always zero. A second application of Rao–Blackwellization removes this limitation. Let ϕ denote the migration rates m, the allele frequencies p and the inbreeding coefficients F. Given ϕ and the data, the ancestries of different individuals are independent: individual k, sampled in population i, has ancestry (s, g) with probability

$$\Pr(z_k = (s, g) \mid \phi, \mathcal{D}) \;\propto\; \pi_i(s, g) \, \Pr(G_k \mid s, g, p, F),$$

where $\pi_i(i,0) = 1 - 3\sum_{j\neq i} m_{ij}$, $\pi_i(s,1) = m_{is}$ and $\pi_i(s,2) = 2m_{is}$ (equations (1)–(3)), and $\Pr(G_k \mid \cdot)$ is the BA3 genotype probability over the observed loci (missing genotypes are omitted, which marginalizes them exactly). With $q_{kj} = \Pr(z_k = (j,1) \mid \phi, \mathcal{D}) + \Pr(z_k = (j,2) \mid \phi, \mathcal{D})$,

$$\Pr(c_{ij} = 0 \mid \phi, \mathcal{D}) = \prod_{k:\,\text{sampled in } i} (1 - q_{kj}), \qquad \Pr(c_{ij} = 0 \mid \mathcal{D}) = \mathbb{E}\Big[\prod_{k}(1 - q_{kj}) \,\Big|\, \mathcal{D}\Big]. \tag{18}$$

With T evaluated states $\phi^{(1)}, \ldots, \phi^{(T)}$ and $w_{ij}^{(t)} = \prod_k (1 - q_{kj}^{(t)})$,

$$\widetilde{\Pr}(c_{ij} = 0 \mid \mathcal{D}) = \frac{1}{T}\sum_{t=1}^{T} w_{ij}^{(t)}, \qquad \widetilde{\mathrm{BF}}_{01} = \widetilde{\Pr}(c_{ij} = 0 \mid \mathcal{D})\,\frac{n_i + K - 1}{K - 1}.$$

Every $w_{ij}^{(t)}$ is positive, so the estimate is positive even when no recorded state has $c_{ij} = 0$. Both estimators are unbiased for the same quantity; the new one replaces each indicator by its conditional expectation given ϕ, which cannot increase the variance.

The product in (18) must be formed state by state. Using instead the posterior ancestry probabilities averaged over the run (the `BA3indiv.txt` values) ignores the dependence between individuals that runs through m: in a simulated data set with 15 individuals per population it understated $\Pr(c_{ij} = 0 \mid \mathcal{D})$ about 20-fold on a true route.

6.2 The collapsed sampler

With `-c` the allele frequencies are not part of the state. Given the ancestries and the latent identity-by-descent and phase codes, the gene-copy counts $n_{sla}^{(t)}$ are sufficient, and the conditional distribution of p_{sl} is Dirichlet$(\alpha_{sla} + n_{sla}^{(t)})$, with $\alpha_{sla} = 1$ under the flat prior and $\alpha_{sla} = c_s \pi_{la}$ under `-H`. At each evaluation BA3 draws $p^{(t)}$ from this distribution, so that $(z^{(t)}, p^{(t)}, m^{(t)}, F^{(t)})$ is a draw from the joint posterior, and then evaluates $w_{ij}^{(t)}$. The draws use a separate random-number stream, so `-B` does not change the chain or any other output.

6.3 Computation

At each evaluation BA3 computes, for every individual and every one of the $2K - 1$ ancestries, the probability of its genotypes. Per population, locus and allele it first tabulates p_{sla} and the homozygote probability $F_s p_{sla} + (1 - F_s) p_{sla}^2$; heterozygotes of non-migrant and first-generation ancestry have probability $2(1 - F_s) p_{sla} p_{slb}$, and second-generation ancestry from s has $p_{sla} p_{ilb}$ for a homozygote and $p_{sla} p_{ilb} + p_{slb} p_{ila}$ for a heterozygote. These are multiplied across loci on the probability scale, with the running product moved to a log offset before it can underflow, so most loci need no logarithm. Then $\log(1 - q_{kj})$ is computed as the log-sum-exp over the ancestries not from j minus the log-sum-exp over all ancestries, which stays accurate when q_{kj} is close to 1, and the averages over states are accumulated in log space. With 4,383 individuals, 2,000 loci and 8 populations one evaluation takes about 0.14 s, so `-B` K evaluates only every K-th recorded state.

6.4 Monte Carlo error

No standard error is reported. When $\Pr(c_{ij} = 0 \mid \mathcal{D})$ is extremely small, $w_{ij}^{(t)}$ varies over many orders of magnitude and a few states dominate the average. The usual estimate of the relative standard error, $\sqrt{(\overline{w^2}/\bar{w}^2 - 1)/T}$, then approaches 1 whatever the data, because $\overline{w^2}/\bar{w}^2 \approx T$ when one term dominates; and it assumes the evaluated states are independent. In this regime $\log_{10} \mathrm{BF}_{01}$ varies between runs in its last digits (-9407 and -9485 for two seeds on a simulated data set with 4,383 individuals) while being far beyond any decision threshold. Where the value is close to a threshold, $w_{ij}^{(t)}$ varies much less, and its precision is best judged by comparing runs with different seeds.

6.5 Validation

On the small exact data sets of Section 11, the estimate agrees with the exact $\Pr(c_{ij} = 0 \mid \mathcal{D})$ for the standard sampler, `-c` and `-c -H` (16 + 4 comparisons, four seeds each, largest $|z| = 2.3$). On a simulated data set with 15 individuals per population and $F_{\mathrm{ST}} = 0.01$ it agrees with the counting estimator on every route where that is nonzero (for example 0.0259 and 0.0255 on a true route), and on a true route where no recorded state had $c_{ij} = 0$ it gives $\log_{10} \mathrm{BF}_{01} \approx -14.6$ and -15.0 for two seeds, where the counting estimator could only report < 0.001.

7 Kullback–Leibler information gain

The `KL(bits)` column reports

$$\mathrm{KL}\big(p(m_{ij} \mid \mathcal{D}) \,\|\, p(m_{ij})\big) = \int_0^{1/3} p(m_{ij} \mid \mathcal{D}) \, \log_2 \frac{p(m_{ij} \mid \mathcal{D})}{p(m_{ij})} \, dm_{ij},$$

the information gained about m_{ij}, with the exact marginal prior (7). The posterior density is estimated from the stored samples of m_{ij} by a Gaussian kernel density estimate, reflected at 0:

$$\hat{p}(m \mid \mathcal{D}) = \frac{1}{Nh\sqrt{2\pi}} \sum_{t=1}^{N} \left[\exp\Big(-\frac{(m - m^{(t)})^2}{2h^2} \Big) + \exp\Big(-\frac{(m + m^{(t)})^2}{2h^2} \Big) \right],$$

with Silverman's bandwidth $h = 1.06 \, \hat{\sigma} \, N^{-1/5}$, where $\hat{\sigma}$ is the posterior standard deviation of m_{ij}. If $h < 10^{-6}$, $h = 0.01$ is used instead. The integral is approximated by the midpoint rule on $G = 1000$ intervals of $[0, \frac{1}{3}]$. Grid points where $\hat{p} < 10^{-10}$ are skipped, and the result is converted to bits and truncated at 0. The KL statistic is descriptive only and is not used in the Bayes factor. Unlike BF_{01}, it depends on a bandwidth and on how well a kernel estimate represents a posterior piled up against 0.

8 Output and interpretation

Each route is reported on one line:

Column	Meaning
`Source`, `Dest`	population indices j and i
`Mean(SD)`	posterior mean and SD of m_{ij}
`Pr(c=0)`	$\widehat{\Pr}(c_{ij} = 0 \mid \mathcal{D})$
`BF_01`	$\widehat{\mathrm{BF}}_{01}$ from (15); `<` marks the convention of Section 5
`log10(BF)`	$\log_{10} \widehat{\mathrm{BF}}_{01}$
`KL(bits)`	information gain, Section 7
`Interpretation`	Jeffreys scale, Table 2

A footer gives the formula, and the range of the bound (16) over the populations in the data set.

With `-B`, a second table follows, headed "Exact Savage-Dickey Test for Zero Migration" and stating how many states were evaluated:

Column	Meaning
`Source`, `Dest`	population indices j and i
`Pr(c=0)`	$\widetilde{\Pr}(c_{ij} = 0 \mid \mathcal{D})$ of Section 6; `~0` when below 10^{-300}
`BF_01`	$\widetilde{\mathrm{BF}}_{01}$, printed as `1e`$k$ when below 10^{-300}
`log10(BF)`	$\log_{10} \widetilde{\mathrm{BF}}_{01}$
`Interpretation`	Jeffreys scale, Table 2

Where both tables give a value, they agree within Monte Carlo error.

$\log_{10} \mathrm{BF}_{01}$	BF_{01}	Interpretation
> 2	> 100	Decisive for H_0 (no migration)
1 to 2	10 to 100	Strong for H_0
0.5 to 1	3.2 to 10	Substantial for H_0
0 to 0.5	1 to 3.2	Weak for H_0
-0.5 to 0	0.32 to 1	Weak for H_1 (migration)
-1 to -0.5	0.1 to 0.32	Substantial for H_1
-2 to -1	0.01 to 0.1	Strong for H_1
< -2	< 0.01	Decisive for H_1

Table 2: Jeffreys scale (Jeffreys, 1961) as used in the output. Because of (16), the categories for H_0 above "Substantial" may be unattainable for small samples (Table 1).

9 Algorithm

Algorithm 1 Savage–Dickey Bayes factors and KL information gain

for each route (i, j), $j \neq i$ **do**
  sumM $\leftarrow 0$, sumM2 $\leftarrow 0$, countZero $\leftarrow 0$, $N \leftarrow 0$, samples $\leftarrow$ empty
end for
for each iteration $t >$ burn-in with $t \bmod s = 0$ **do**
  for each route (i, j), $j \neq i$ **do**
    $m \leftarrow m_{ij}$; sumM $\leftarrow$ sumM $+ m$; sumM2 $\leftarrow$ sumM2 $+ m^2$; append m to samples
    if `migrantCounts[i][j][1]` + `migrantCounts[i][j][2]` $= 0$ **then**
      countZero $\leftarrow$ countZero $+ 1$
    end if
    $N \leftarrow N + 1$
  end for
end for
for each destination i **do**
  $n_i \leftarrow \sum_j \sum_{a=0}^{2}$ `migrantCounts[i][j][a]` ▷ individuals sampled in i
  $B_i \leftarrow (n_i + K - 1)/(K - 1)$ ▷ largest attainable BF_{01}
  for each source $j \neq i$ **do**
    if $N < 100$ **then**
      report "Insufficient samples"; **continue**
    end if
    $\hat{p} \leftarrow$ countZero$/N$
    if countZero $= 0$ **then**
      $\mathrm{BF}_{01} \leftarrow B_i/N$, marked "<"
    else
      $\mathrm{BF}_{01} \leftarrow \hat{p}\, B_i$
    end if
    $\hat{\sigma} \leftarrow \sqrt{\mathrm{sumM2}/N - (\mathrm{sumM}/N)^2}$; $h \leftarrow 1.06\, \hat{\sigma}\, N^{-1/5}$; **if** $h < 10^{-6}$ **then** $h \leftarrow 0.01$
    KL $\leftarrow \sum_{g=0}^{G-1} \hat{p}(m_g) \log_2\big(\hat{p}(m_g)/p(m_g)\big)\, \Delta m$, $m_g = (g + \frac{1}{2})\Delta m$, $\Delta m = \frac{1}{3G}$, $p(m)$ from (7)
    report mean, SD, $\hat{p}$, BF_{01}, $\log_{10} \mathrm{BF}_{01}$, $\max(\mathrm{KL}, 0)$, interpretation
  end for
end for

Algorithm 2 Exact Bayes factors (`-B` K), one evaluation

if collapsed sampler **then**
  draw $p_{sl} \sim \text{Dirichlet}(\alpha_{sla} + n_{sla})$ for every population s and locus l ▷ own RNG stream
else
  take p from the current state
end if
tabulate p_{sla} and $F_s p_{sla} + (1 - F_s) p_{sla}^2$ for every s, l, a
for each individual k, sampled in population i **do**
  $\ell_s^{\text{pure}} \leftarrow \log \Pr(G_k \mid \text{both genes from } s, F_s)$ for every s ▷ observed loci only
  $\ell_s^{\text{hyb}} \leftarrow \log \Pr(G_k \mid \text{one gene from } s, \text{one from } i)$ for every $s \neq i$
  $c_0 \leftarrow \log(1 - 3 \sum_{j \neq i} m_{ij}) + \ell_i^{\text{pure}}$
  $c_{1s} \leftarrow \log m_{is} + \ell_s^{\text{pure}}$ and $c_{2s} \leftarrow \log 2m_{is} + \ell_s^{\text{hyb}}$ for every $s \neq i$
  $T \leftarrow \text{logsumexp}(c_0, c_{1s}, c_{2s} : s \neq i)$
  for each source $j \neq i$ **do**
    $A_{ij} \leftarrow A_{ij} + \text{logsumexp}(c_0, c_{1s}, c_{2s} : s \neq i, j) - T$ ▷ $\log(1 - q_{kj})$
  end for
end for
for every route: $S_{ij} \leftarrow \text{logsumexp}(S_{ij}, A_{ij})$; reset $A_{ij} \leftarrow 0$; $T_{\text{eval}} \leftarrow T_{\text{eval}} + 1$
at the end: $\log \widetilde{\Pr}(c_{ij} = 0 \mid \mathcal{D}) = S_{ij} - \log T_{\text{eval}}$; $\log \widetilde{\text{BF}}_{01} = \log \widetilde{\Pr} + \log \frac{n_i + K - 1}{K - 1}$

10 Implementation

The code is in `src/main.cpp` (functions `initSavageDickeyStats`, `updateSavageDickeyStats`, `computeKDEatPoint`, `computeKLDivergence` and `computeSavageDickeyBayesFactors`), with the statistics held in

```
struct SavageDickeyStats {
    double sumM;                 // running sum of m_ij (mean)
    double sumM2;                // running sum of m_ij^2 (SD)
    long int countZero;          // recorded states with c_ij = 0
    long int nSamples;           // recorded states N
    std::vector<double> samples; // m_ij samples, for the KL estimate
};
```

`updateSavageDickeyStats` is called in the sampling block of the main loop, after burn-in and every `-n` iterations, with the current migration rates and ancestry counts. Storing the samples for the KL estimate needs $O(NK^2)$ memory. The Bayes factor itself needs only the counter `countZero`.

The `-B` estimator is in `struct ExactBF` and the functions `exactBFInit`, `exactBFStep`, `exactBFReport` and `exactBFFree`. `exactBFStep` is called next to `updateSavageDickeyStats`, on every K-th recorded state. The structure holds, per route, the log of the running sum of $w_{ij}^{(t)}$ and the current evaluation's $\sum_k \log(1 - q_{kj})$; a mask of observed genotypes; the allele and homozygote probabilities of the current evaluation, stored contiguously by population, locus and allele; and its own random-number generator, seeded from the run's seed.

11 Validation

`tests/exact_posterior.py` computes the exact BA3 posterior for tiny data sets by enumerating all $(2K - 1)^N$ ancestry assignments, with m, the allele frequencies and F integrated out. It reports the exact $\Pr(c_{ij} = 0 \mid \mathcal{D})$ and BF_{01} from (15); `tests/exact_posterior_hier.py` does the same under `-H`. `tests/check_exact.py` runs BA3 with several seeds and compares

the reported `Pr(c=0)` and `BF_01` (and all other posterior summaries) with the exact values by z-scores. When the change was made, the standard sampler, `-c` and `-c -H` all passed (largest $|z| = 2.1$; commit `52be5a1`). The test compares BA3's estimate with the exact $\Pr(c_{ij} = 0 \mid \mathcal{D})$, so it checks the sampling of ancestries and the reported numbers. The derivation of (15) from (10) is checked by the two simulations of Sections 1 and 4, which run BA3's migration move directly. The `-B` estimator was checked against the same exact values (Section 6); with `-B` off, the output of the standard sampler is byte-identical to 3.6.1 (`tests/regress_default.sh`, 21 of 21).

12 Versions 3.4.0–3.5.0 (superseded)

Versions 3.4.0 to 3.5.0 estimated the posterior density at zero from the samples of m_{ij} by a Gaussian kernel density estimate, reflected at 0, and divided it by a prior density of 3. The method has two errors.

1. **Inconsistent bandwidth.** The kernel sum $S = \sum_t \exp\big(-(m^{(t)})^2/2h_{\text{ref}}^2\big)$ was accumulated during the run with a fixed $h_{\text{ref}} = 0.02$, but normalized at the end with Silverman's bandwidth h_{opt}: $\hat{p}(0 \mid \mathcal{D}) = 2S/(Nh_{\text{opt}}\sqrt{2\pi})$. When the posterior is concentrated near zero, $h_{\text{opt}} \ll h_{\text{ref}}$ and the density at zero is overstated by roughly $h_{\text{ref}}/h_{\text{opt}}$. A histogram estimate, the fraction of samples with $m_{ij} < 0.005$ divided by 0.005, was substituted only when the two differed by more than a factor of 100.

2. **Wrong prior density.** The prior density at zero was taken as 3, as if each m_{ij} were Uniform$(0, \frac{1}{3})$. The prior that BA3 samples is (6), with density $3(K-1)$ at zero (8). For $K > 2$ this overstated BF_{01} by a further factor of $K - 1$, and the histogram fallback had the same error.

Together these overstated BF_{01} by a factor of about 30 in a 10-population example (commit `52be5a1`). Evidence for H_0 reported by these versions, especially "Strong" or "Decisive" for no migration, should be recomputed with BA3 3.6.0 or later. The KL column of these versions used the same Uniform$(0, \frac{1}{3})$ prior. Sampling was not affected: with the Bayes-factor table excluded, 3.6.0 output is byte-identical to 3.5.0 for the same seed (`tests/regress_default.sh --no-sd`).

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